Regularisation and early stopping

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Regularisation and early stopping before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Training and validation curves

Without opening the old lesson, explain one key idea from **Training** and validation curves. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Batching and epochs

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Gradients with a numerical check

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Keep h = ReLU(x), y = vh, train only v on (1, 2) and (2, 4), and freeze all hidden parameters and the zero **output** bias. **Training** MSE is L(v) = 2.5(v - 2) squared.

Define J(v) = L(v) + lambda times v squared, with lambda nonnegative. Only this one **output** weight is penalised here.

For lambda = 0.5, the penalty at v = 1.5 is 0.5 times 2.25 = 1.125. **Prediction** MSE is 0.625, so total objective J is 1.75. Keep these quantities separate in logs.

A smaller total objective does not automatically mean a smaller **prediction** error.

v	Prediction MSE	Penalty at lambda 0.5	Total J
1	2.5	0.5	3
1.5	0.625	1.125	1.75
2	0	2	2

Among these three candidates, the smallest penalised objective is at v = 1.5, whereas the smallest **training** MSE is at v = 2. Penalising a weight changes the tradeoff.

It does not guarantee better validation performance or fair outcomes.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Regularisation and early stopping trace and record the first step you can explain without help.

Use W2: Trace the next update for **Regularisation** and early stopping. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Regularisation and early stopping explanation against the supplied facts before discussing it.

Use the worked case for **Regularisation** and early stopping. Proposed explanation: “Zero **prediction** error means no regularised update”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Transfer both rules independently

For a new objective J(a) = (a - 2) squared + a squared, gradient 4a - 4, start a = 0 and take one step at rate 0.25. Give new a, **prediction** error term, penalty and total J.

Separately apply **patience** 2 with strict improvement to validation values [3, 2, 2, 2, 1] at epochs 0 to 4; report the stop and **restore**d epochs.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Separate the objective terms

> Separate the objective terms. Apply the supplied Regularisation and early stopping facts and keep your reasoning.

At v = 2 and lambda = 0.5 in the lesson, compute **prediction** MSE, penalty, total objective and total gradient. Does zero **training** MSE make this gradient zero?

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the next update

> Trace the next update. Apply the supplied Regularisation and early stopping facts and keep your reasoning.

Start v = 1.6 with lambda = 0.5 and rate 0.1. Calculate both gradient contributions, new v and new **prediction** MSE.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Apply patience without peeking

> Apply patience without peeking. Apply the supplied Regularisation and early stopping facts and keep your reasoning.

Validation MSE at epochs 0 to 7 is [4, 2, 1.5, 1.6, 1.4, 1.45, 1.46, 1.3]. Apply the lesson policy with **patience** 2. Give the stop epoch, **restore**d epoch and why the listed epoch 7 cannot change this decision.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Transfer both rules independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer both rules independently. Apply the supplied Regularisation and early stopping facts and keep your reasoning.

For a new objective J(a) = (a - 2) squared + a squared, gradient 4a - 4, start a = 0 and take one step at rate 0.25. Give new a, **prediction** error term, penalty and total J.

Separately apply **patience** 2 with strict improvement to validation values [3, 2, 2, 2, 1] at epochs 0 to 4; report the stop and **restore**d epochs.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Compare penalties fairly

> Compare penalties fairly. Apply the supplied Regularisation and early stopping facts and keep your reasoning.

A team compares lambda 0 and lambda 0.5 by choosing whichever has the smaller penalised **training** objective. Explain why this does not establish the better **prediction** model and name a suitable development comparison.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Regularisation and early stopping to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled **regularisation** and early stopping **evidence** sheet to Image-pattern Network Investigation. Show how you can separate a **weight penalty** from **prediction** error.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Regularisation and early stopping

> Separate a regularisation penalty from prediction loss and an early-stopping rule. Explain the labelled step before checking the changed condition.

Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule.

Toy objective $J(a)=(a-3)^2+2a^2$; start a = 0; gradient 6a-6; rate 1/6. Separate validation losses:[5, 3, 3, 4], **patience** 1, no ties as improvement.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/16

Regularisation and early stopping

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Regularised update

At 0 gradient -6; update gives a = 1.

2. Split objective

At 1 prediction term 4, penalty 2, total 6; at 0 total 9.

3. Early-stop separately

Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

CHANGE ONE THING

Change patience from 1 to2 under the same tie rule.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

1. Split objective

2. Early-stop separately

3. Regularised update

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Early-stop separately Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

Regularisation adds a parameter penalty to the **training** objective. Here it discourages large a, so the objective balances **prediction** error against that penalty.

Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the **patience** allowance is used and **restore** the best checkpoint.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change **patience** from 1 to2 under the same tie rule. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Stop at checkpoint 4; still **restore** checkpoint 2.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Stop at checkpoint 4; still **restore** checkpoint 2. Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

Repair a missing building block

Review **Training** and validation curves (ai-5/15). Goal: Compute comparable **training** and validation losses at saved checkpoints. Short practice: Can the original epoch 3 **training** loss 1 justify claiming best generalisation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Batching and epochs (ai-5/13). Goal: Distinguish an example a batch an update and an epoch. Short practice: Why do the single-example orders finish at different weights?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Gradients with a numerical check (ai-5/10). Goal: Estimate one partial derivative while other parameters stay fixed. Short practice: Why should the numerical checks perturb one coordinate at a time?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

Explain each Regularisation and early stopping picture from its labels, then record a relevant limit.

01. Regularisation and early stopping: the case

AI ACADEMY / AI-5/16 · 01 · LOOK AT THE FACTS

Regularisation and early stopping: the case

The situation

Separate a regularisation penalty from prediction loss and an early-stopping rule.

Supplied evidence

Toy objective $J(a)=(a-3)^2+2a^2$; start $a = 0$; gradient $6a-6$; rate $1/6$. Separate validation losses:[5, 3, 3, 4], patience 1, no ties as improvement.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain regularisation and early stopping using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

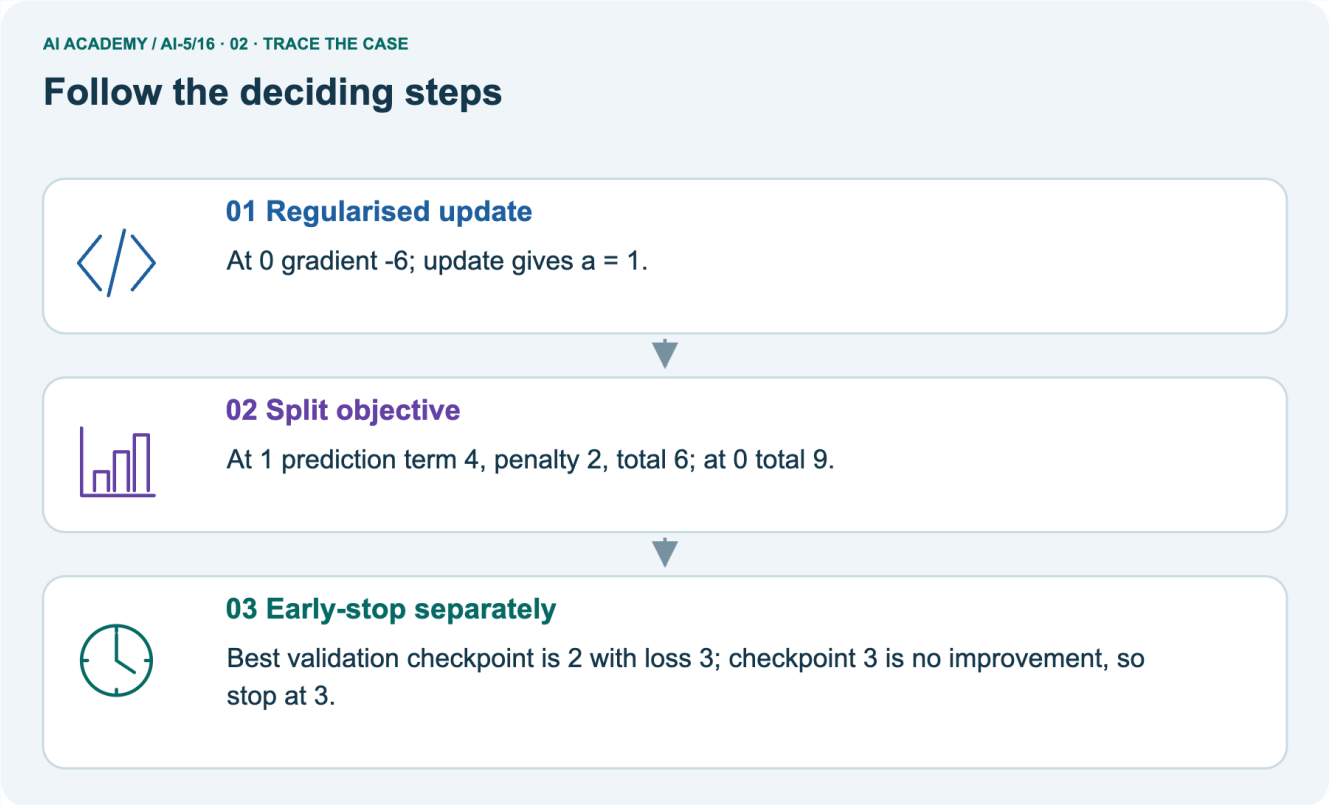
Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule. Supplied facts: Toy objective $J(a)=(a-3)^2+2a^2$; start $a = 0$; gradient $6a-6$; rate $1/6$.

Separate validation losses:[5, 3, 3, 4], **patience** 1, no ties as improvement. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Regularised update: At 0 gradient -6; update gives a = 1. → Split objective: At 1 **prediction** term 4, penalty 2, total 6; at 0 total 9. → Early-stop separately: Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Patience two with strict improvement

AI ACADEMY / AI-5/16 · 03 · INSPECT THE LESSON MODEL

Patience two with strict improvement

Epoch	Validation MSE	Decision
0 then 1 then 2	4 then 2 then 1.5	Save each improved state
3 then 4	1.6 then 1.4	Wait once then reset
5 then 6	1.45 then 1.46	Two misses so stop at 6

Initial checkpoint is epoch 0. Evaluate once after each epoch. A strict decrease is improvement; an exact tie is a miss. Reset the miss count after improvement and stop at two consecutive misses. Restore epoch 4, the best observed checkpoint.

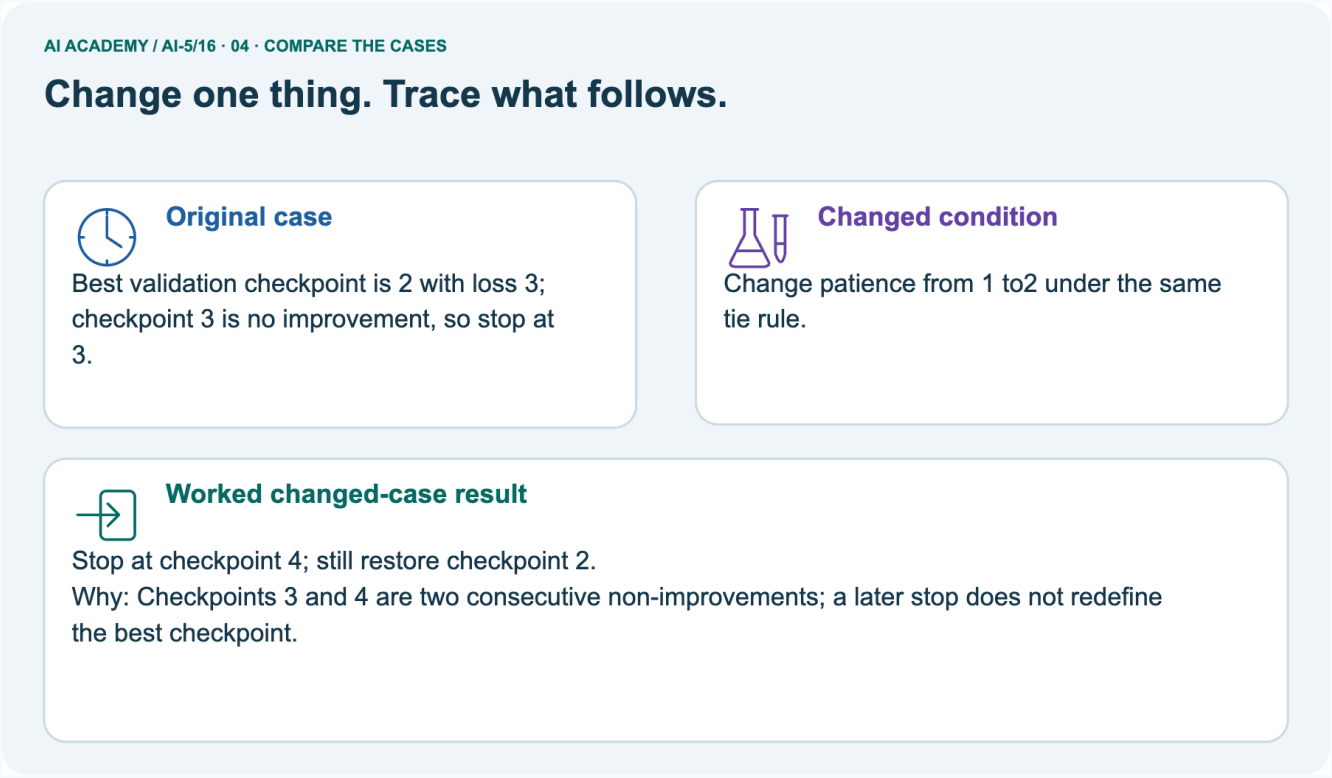
Follow the labelled picture with the matching explanation. Open the vector diagram

Patience two with strict improvement. 0 then 1 then 2 | 4 then 2 then 1.5 | Save each improved state. 3 then 4 | 1.6 then 1.4 | Wait once then reset. 5 then 6 | 1.45 then 1.46 | Two misses so stop at 6

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

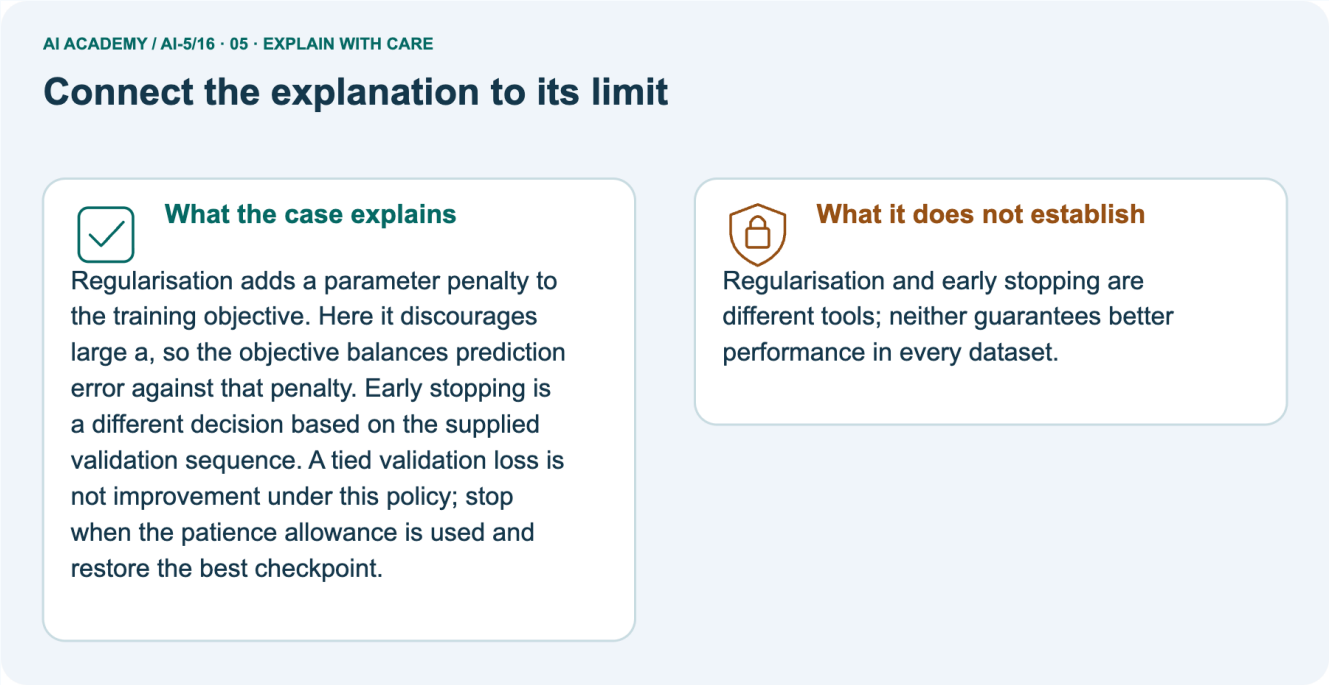
Original case: Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3. Changed condition: Change **patience** from 1 to2 under the same tie rule.

Expected result: Stop at checkpoint 4; still **restore** checkpoint 2. Reason: Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Regularisation adds a parameter penalty to the **training** objective. Here it discourages large a, so the objective balances **prediction** error against that penalty.

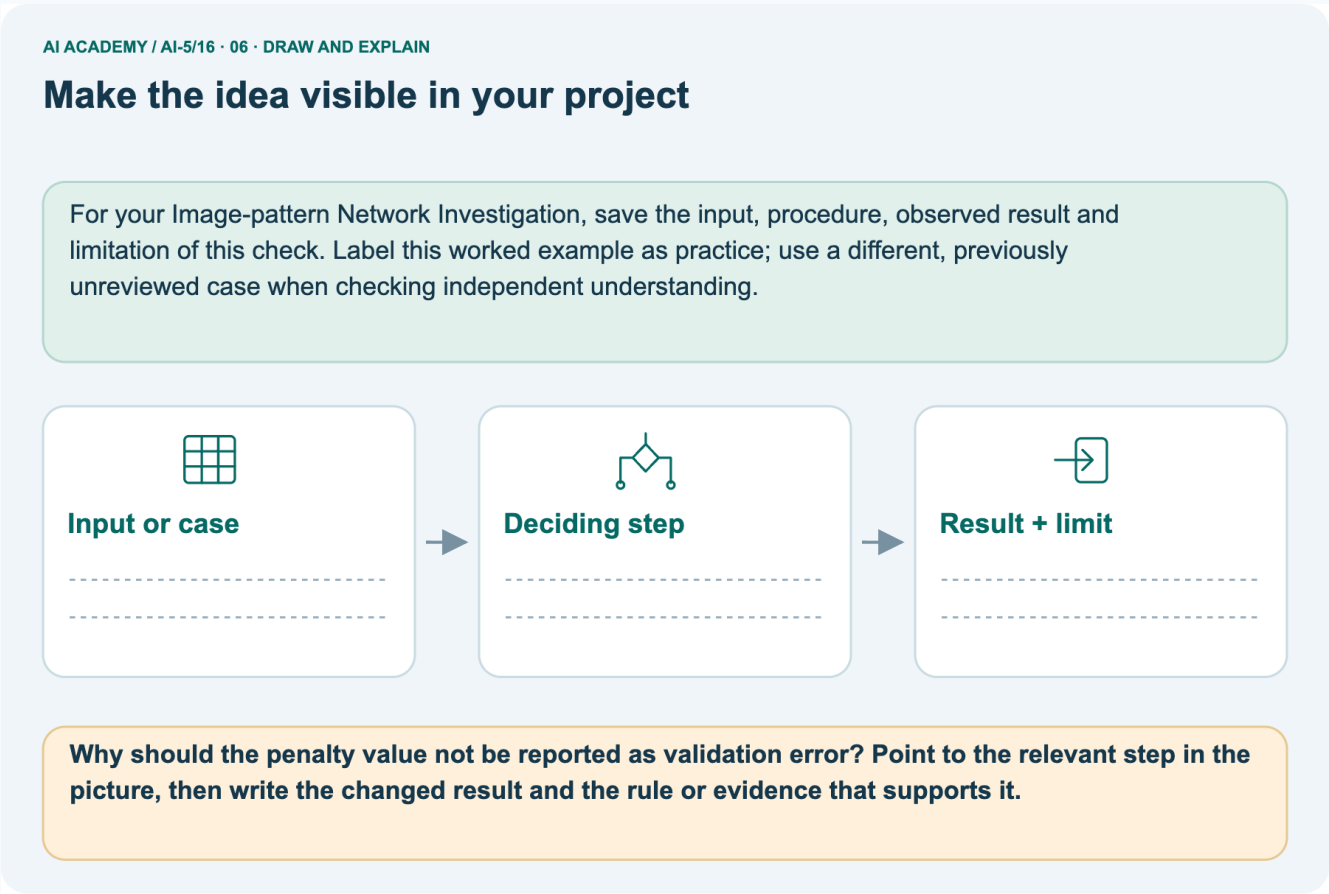
Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the **patience** allowance is used and **restore** the best checkpoint.

Limit: **Regularisation** and early stopping are different tools; neither guarantees better performance in every dataset.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.